

Teaching Statement

Regina (Jeanne) Ruane

Teaching lies at the core of my academic identity. Across two decades in higher education—spanning research universities, engineering schools, and business programs—I have sought to design and deliver pedagogical experiences that equip students not only with technical proficiency, but also with the capacity to think critically, reason ethically, and apply data-driven insights to real-world problems.

My teaching philosophy is rooted in three pillars: (1) interdisciplinary integration, (2) research-informed instruction, and (3) inclusive and adaptive pedagogy.

First, I center my teaching around interdisciplinary integration. With appointments spanning data science, engineering, business, and education, I have consistently designed curricula that bridge disciplinary boundaries. I currently teach courses such as Sample Survey Methods (Penn), Statistical Learning and Data Mining, Social Network Analysis and Business Analytics, and Information Economics (Temple). In these courses, I embed core concepts from statistics, machine learning, and network science within application-rich contexts ranging from online education to FinTech and public policy. My goal is to demonstrate how data science can illuminate complex systems—be they social, technical, or economic—and to empower students to develop models that are both analytically rigorous and socially meaningful.

Second, my teaching is research-informed and experiential. I incorporate live coding in R and Python, bring anonymized real-world datasets into the classroom, and routinely align class projects with my own applied research. For example, students in my Social Network Analysis course have worked on digital interaction data from city policy interventions, analyzing social diffusion and behavior change. In collaboration with industry partners like Wells Fargo and the City of Philadelphia, I have also created project-based learning modules that engage students in modeling strategies relevant to practice. These experiences develop both technical fluency and the ability to communicate results to stakeholders across domains.

Third, I maintain a strong commitment to inclusive and adaptive pedagogy. I have taught students from a wide range of backgrounds—undergraduate to doctoral level, traditional and executive formats, in-person, hybrid, and online. I approach this diversity as an asset and use backward design and formative assessment to scaffold learning across ability levels. As the former founding director of Drexel's Exploratory Studies Program, I built a peer mentoring system and advising model that increased student retention and doubled enrollment. I continue to use these insights to create learning

environments that are accessible, participatory, and equity-minded. I also lead efforts around women in data science and actively mentor students pursuing interdisciplinary, applied research.

Looking forward, I am eager to contribute to Princeton's data science initiative by designing courses that integrate core methodological tools with applications in public policy, education, financial systems, and AI governance. I am particularly excited about developing interdisciplinary project studios and thesis mentoring models that encourage students to translate abstract methods into actionable insights. I see the classroom not just as a site of technical training, but as a collaborative, critical space where students learn to question the world around them, work across disciplines, and build the tools to shape a more informed, equitable future.